

Multi-Task Graph Neural Networks for Political Tweets

Political Misinformation Social Network Project

Overview

This project represents political tweets as a social graph and evaluates graph neural network (GNN) architectures. The model has two prediction heads: one predicts the political account that authored a tweet, and the other predicts whether the tweet received a generated misinformation label.

Dataset and graph construction

The working archive contains 7,053 tweets from 14 political accounts. The graph contains 10,160 nodes and 97,364 edges after adding social-network entities from the tweet metadata. Only tweet nodes are labeled; the remaining graph nodes provide neighborhood context.

Labels

Misinformation labels were generated with joeddav/xlm-roberta-large-xnli using valid, misleading, and invalid. These are pseudo-labels rather than expert fact-checks, so the results should not be interpreted as a definitive misinformation detector.

Models

Each model uses two graph convolution layers, normalization, ReLU activations, dropout, and a shared hidden representation. Two linear heads predict the 14-class author label and the binary misinformation label. The comparison includes GCN, GraphSAGE, and GAT.

Results

Model	Author acc.	Author F1	Misinfo acc.	Misinfo F1
GCN	0.962	0.957	0.755	0.689
GraphSAGE	0.991	0.989	0.775	0.708
GAT	0.997	0.997	0.703	0.641

GraphSAGE gives the best balance between the two objectives. GAT achieves the highest author-identification score, but its misinformation macro-F1 is lower. This suggests that identifying an account is easier than recovering the noisier generated labels.

Conclusion

The experiment demonstrates a practical multi-task GNN workflow on a medium-sized political social graph. A stronger follow-up would replace zero-shot labels with independently verified fact-check annotations and evaluate across time-based splits.

Code

github.com/VikiShmiki/Political-Misinformation-Social-Networks